C

ACT MATH • UNIT 0.01

ACT Math Diagnostic & Pacing Strategy Drill

ANSWER KEY & SOLUTIONS

60-Minute Pacing & Core Readiness • Official ACT Math Curriculum

Total Questions: 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 C	Q3 B	Q4 A	Q5 A
Q6 D	Q7 A	Q8 D	Q9 B	Q10 C
Q11 D	Q12 A	Q13 B	Q14 A	Q15 B
Q16 C	Q17 D	Q18 B	Q19 D	Q20 D
Q21 B	Q22 A	Q23 B	Q24 A	Q25 B

ACT Math Diagnostic & Pacing Strategy Drill Practice Exercises

1 MCQ

For the imaginary unit i, where $i^2 = -1$, which of the following complex numbers is equal to $(5 - 3i) - (2 - 8i)$?

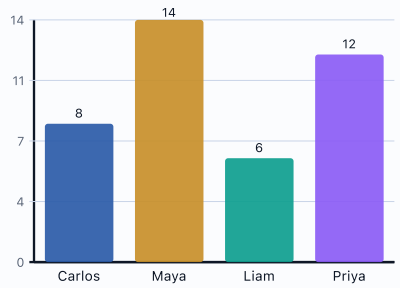
- **A** $3 + 5i$ ✓
- **B** $3 - 11i$
- **C** $7 - 11i$
- **D** $3 - 5i$

WORKED SOLUTION & VERIFICATION

To subtract complex numbers, distribute the negative sign to both terms inside the second set of parentheses:
$(5 - 3i) - (2 - 8i) = 5 - 3i - 2 + 8i$
Next, combine the real parts and the imaginary parts separately:
Real parts: $5 - 2 = 3$
Imaginary parts: $(-3 + 8)i = 5i$
Combining these yields $3 + 5i$.
Common Misconceptions:
- $3 - 11i$ occurs if the negative sign is not distributed to $-8i$, yielding $-3i - 8i = -11i$.
- $7 - 11i$ occurs if the real parts are added $(5 + 2 = 7)$ instead of subtracted.
- $3 - 5i$ occurs from an arithmetic sign error when combining $-3 + 8$.

2 MCQ

The bar graph below shows the number of hours four students volunteered at a community center last week. What percent of the total volunteer hours logged by the four students was logged by Maya?



A 14%

B 28%

C 35% ✓

D 42%

WORKED SOLUTION & VERIFICATION

To find the percentage of total hours logged by Maya:

1. Find the total number of volunteer hours logged by all four students:

$$\text{Total hours} = 8 + 14 + 6 + 12 = 40$$

2. Determine the fraction of hours Maya contributed:

$$\frac{\text{Maya's hours}}{\text{Total hours}} = \frac{14}{40}$$

3. Simplify the fraction and convert to a percentage:

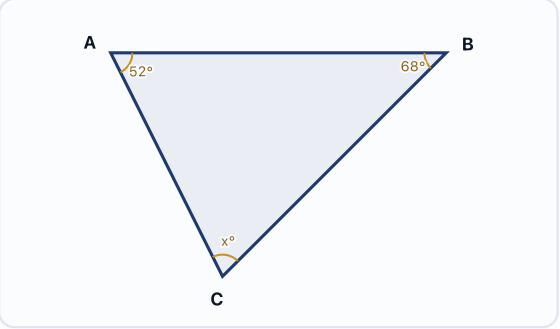
$$\frac{14}{40} = \frac{7}{20} = \frac{35}{100} = 35\%$$

Common Errors:

- Choosing 14% mistakenly treats the raw number of hours as a percentage without dividing by the total.
- Choosing 28% corresponds to calculating $\frac{14}{50}$ instead of using the actual total of 40 hours.

3 MCQ

In $\triangle ABC$ shown below, the measure of $\angle A$ is 52° and the measure of $\angle B$ is 68°. What is the value of x?



A 58

B 60 ✓

C 70

D 120

WORKED SOLUTION & VERIFICATION

The sum of the interior angle measures in any triangle is 180°.

Set up the equation for the sum of the angles in $\triangle ABC$:

$$\angle A + \angle B + \angle C = 180^\circ$$

Substitute the given values into the equation:

$$52^\circ + 68^\circ + x^\circ = 180^\circ$$

Combine the known angle measures:

$$120^\circ + x^\circ = 180^\circ$$

Subtract 120° from both sides:

$$x = 180 - 120 = 60$$

- A common error is calculating $52 + 68 = 120$ and stopping there, which gives 120.
- Another mistake is an arithmetic error such as $180 - 110 = 70$.

4 MCQ

In the right triangle $\triangle ABC$ shown below, the right angle is at vertex C. The lengths of the sides are given in inches: $AC = 12$, $BC = 5$, and $AB = 13$. What is the value of $\cos B$?



A $\frac{5}{13}$ ✓

B $\frac{12}{13}$

C $\frac{5}{12}$

D $\frac{12}{5}$

WORKED SOLUTION & VERIFICATION

In any right triangle, the cosine of an acute angle is defined as the ratio of the length of the side adjacent to the angle to the length of the hypotenuse:

$$\cos B = \frac{\text{length of side adjacent to } B}{\text{length of hypotenuse}} = \frac{BC}{AB}$$

From the given triangle:

- The side adjacent to $\angle B$ is $BC = 5$.
- The hypotenuse is $AB = 13$.

Therefore, $\cos B = \frac{5}{13}$.

Common Misconceptions:

- $\frac{12}{13}$ represents $\sin B = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{AC}{AB}$ or $\cos A$.
- $\frac{5}{12}$ represents $\frac{\text{adjacent}}{\text{opposite}} = \cot B$ (or $\tan A$).
- $\frac{12}{5}$ represents $\tan B = \frac{\text{opposite}}{\text{adjacent}} = \frac{AC}{BC}$.

5 MCQ

In the standard (x, y) coordinate plane, vector $\mathbf{u} = \langle 2, -5 \rangle$ and vector $\mathbf{v} = \langle k, 3 \rangle$, where $k > 0$. If the magnitude of the vector sum $\mathbf{u} + \mathbf{v}$ is $\sqrt{29}$, what is the value of k?

- **A** 3 ✓
- **B** 7
- **C** $\sqrt{21}$
- **D** 5

WORKED SOLUTION & VERIFICATION

To find k, follow these steps:

1. **Find the vector sum $\mathbf{u} + \mathbf{v}$:**

Add corresponding components:

$$\mathbf{u} + \mathbf{v} = \langle 2 + k, -5 + 3 \rangle = \langle k + 2, -2 \rangle$$

2. **Express the magnitude of $\mathbf{u} + \mathbf{v}$:**

The magnitude of a vector $\langle x, y \rangle$ is given by $\sqrt{x^2 + y^2}$:

$$\|\mathbf{u} + \mathbf{v}\| = \sqrt{(k + 2)^2 + (-2)^2} = \sqrt{(k + 2)^2 + 4}$$

3. **Set up and solve the equation:**

Given that the magnitude is $\sqrt{29}$:

$$\sqrt{(k + 2)^2 + 4} = \sqrt{29}$$

Square both sides:

$$(k + 2)^2 + 4 = 29$$

$$(k + 2)^2 = 25$$

Taking the square root of both sides gives $k + 2 = \pm 5$. Since $k > 0$, we take the positive root:

$$k + 2 = 5 \implies k = 3$$

Common Traps:

- Choice B (5) results from solving for $k + 2 = 5$ but forgetting to subtract 2 to solve for k.
- Choice C (7) results from a sign error in vector addition, writing the horizontal component as $k - 2$, leading to $k - 2 = 5 \implies k = 7$.
- Choice D ($\sqrt{21}$) results from the common algebraic error of expanding $(k + 2)^2$ as $k^2 + 4$, giving $k^2 + 4 + 4 = 29 \implies k^2 = 21 \implies k = \sqrt{21}$.

6 MCQ

Which of the following expressions is equivalent to $(5x^2 - 2x + 7) - (3x^2 - 6x - 4)$?

A $2x^2 - 8x + 3$

B $2x^2 - 8x + 11$

C $2x^2 + 4x + 3$

D $2x^2 + 4x + 11$ ✓

WORKED SOLUTION & VERIFICATION

To simplify the expression, distribute the negative sign across each term in the second set of parentheses:

$$-(3x^2 - 6x - 4) = -3x^2 + 6x + 4$$

Now rewrite the entire expression and combine like terms:

$$(5x^2 - 2x + 7) - 3x^2 + 6x + 4$$

Group terms with the same power of x:
- Quadratic terms: $5x^2 - 3x^2 = 2x^2$
- Linear terms: $-2x + 6x = 4x$
- Constant terms: $7 + 4 = 11$

Combining these yields $2x^2 + 4x + 11$.

Common misconceptions:
- Failing to distribute the negative sign to $-6x$ yields $-2x - 6x = -8x$, producing the distractor $2x^2 - 8x + 11$.
- Failing to distribute the negative sign to -4 yields $7 - 4 = 3$, producing the distractor $2x^2 + 4x + 3$.
- Failing to distribute the negative sign to both terms yields $2x^2 - 8x + 3$.

7 MCQ

In the standard (x, y) coordinate plane, what are the coordinates of the midpoint of a line segment with endpoints $(-3, 8)$ and $(5, -2)$?

A $(1, 3)$ ✓

B $(4, -5)$

C $(2, 6)$

D $(3, 1)$

WORKED SOLUTION & VERIFICATION

To find the midpoint (x_m, y_m) of a line segment with endpoints (x_1, y_1) and (x_2, y_2), use the midpoint formula:

$$x_m = \frac{x_1 + x_2}{2}, \quad y_m = \frac{y_1 + y_2}{2}$$

Substitute the given coordinates $(-3, 8)$ and $(5, -2)$:

$$x_m = \frac{-3 + 5}{2} = \frac{2}{2} = 1$$

$$y_m = \frac{8 + (-2)}{2} = \frac{6}{2} = 3$$

Thus, the midpoint is $(1, 3)$.

Common errors:
- Finding the difference instead of the average yields $(\frac{5-(-3)}{2}, \frac{-2-8}{2}) = (4, -5)$.
- Forgetting to divide the sum by 2 yields $(2, 6)$.
- Reversing the x- and y-coordinates yields $(3, 1)$.

8 MCQ

If $f(x) = 3x^2 - 5x + 4$, what is the value of $f(-2)$?

A -18

B 2

C 6

D **26** ✓

WORKED SOLUTION & VERIFICATION

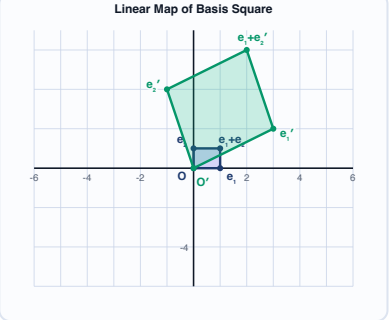
To evaluate $f(-2)$, substitute -2 for x in the function rule: $f(-2) = 3(-2)^2 - 5(-2) + 4$. First, evaluate the exponent: $(-2)^2 = 4$. Next, perform the multiplications: $3(4) = 12$ and $-5(-2) = +10$. Finally, sum the terms: $12 + 10 + 4 = 26$. Common errors include treating $(-2)^2$ as -4, which yields $-12 + 10 + 4 = 2$, or failing to apply the negative sign to the middle term, giving $12 - 10 + 4 = 6$.

9 MCQ

In the standard (x, y) coordinate plane, the complex number $z = x + yi$ satisfies the equation

$$z(2 - i) = 15 + 5i$$

where $i^2 = -1$, and x and y are real numbers. Vector $\mathbf{v}$ is defined by $\mathbf{v} = \begin{bmatrix} x \\ y \end{bmatrix}$. If matrix $A = \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix}$, what is the magnitude of the vector $A\mathbf{v}$?



A $10\sqrt{5}$

B $10\sqrt{10}$ ✓

C 40

D $5\sqrt{2}$

WORKED SOLUTION & VERIFICATION

Step 1: Solve for the complex number $z = x + yi$.

Multiply the numerator and denominator by the complex conjugate of $2 - i$, which is $2 + i$:

$$z = \frac{15 + 5i}{2 - i} = \frac{(15 + 5i)(2 + i)}{(2 - i)(2 + i)} = \frac{30 + 15i + 10i + 5i^2}{4 - i^2}$$

Since $i^2 = -1$:

$$z = \frac{30 + 25i - 5}{4 - (-1)} = \frac{25 + 25i}{5} = 5 + 5i$$

Thus, $x = 5$ and $y = 5$.

Step 2: Construct vector $\mathbf{v}$ and perform the matrix-vector multiplication $A\mathbf{v}$:

$$\mathbf{v} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$$

$$A\mathbf{v} = \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ 5 \end{bmatrix} = \begin{bmatrix} 3(5) + (-1)(5) \\ 2(5) + 4(5) \end{bmatrix} = \begin{bmatrix} 15 - 5 \\ 10 + 20 \end{bmatrix} = \begin{bmatrix} 10 \\ 30 \end{bmatrix}$$

Step 3: Calculate the magnitude of the resulting vector $A\mathbf{v}$:

$$\|A\mathbf{v}\| = \sqrt{10^2 + 30^2} = \sqrt{100 + 900} = \sqrt{1000} = 10\sqrt{10}$$

Common Misconceptions:

- Choice A ($5\sqrt{2}$) is the magnitude of $\mathbf{v}$ alone, forgetting to apply the transformation matrix A.
- Choice D (40) incorrectly sums the components $10 + 30$ instead of computing the Euclidean norm $\sqrt{10^2 + 30^2}$.

10 MCQ

In the standard (x, y) coordinate plane, vector $\mathbf{u} = \langle 2, -5 \rangle$ and vector $\mathbf{v} = \langle -3, 4 \rangle$. What is the vector $2\mathbf{u} + \mathbf{v}$?

A $\langle -1, -1 \rangle$

B $\langle 1, -1 \rangle$

C $\langle 1, -6 \rangle$ ✓

D $\langle 7, -14 \rangle$

WORKED SOLUTION & VERIFICATION

To find $2\mathbf{u} + \mathbf{v}$, first perform the scalar multiplication on $\mathbf{u}$:

$$2\mathbf{u} = 2\langle 2, -5 \rangle = \langle 2(2), 2(-5) \rangle = \langle 4, -10 \rangle$$

Next, add the corresponding components of $2\mathbf{u}$ and $\mathbf{v}$:

$$2\mathbf{u} + \mathbf{v} = \langle 4 + (-3), -10 + 4 \rangle = \langle 1, -6 \rangle$$

Common misconceptions:

- Computing $\mathbf{u} + \mathbf{v}$ without multiplying by 2 yields $\langle 2 + (-3), -5 + 4 \rangle = \langle -1, -1 \rangle$.
- Forgetting to multiply the y-component of $\mathbf{u}$ by 2 yields $\langle 4 + (-3), -5 + 4 \rangle = \langle 1, -1 \rangle$.
- Subtracting $\mathbf{v}$ instead of adding yields $\langle 4 - (-3), -10 - 4 \rangle = \langle 7, -14 \rangle$.

11 MCQ

A bag contains 8 red marbles, 6 green marbles, and 10 blue marbles. If 1 marble is chosen at random from the bag, what is the probability that the marble is NOT red?

A $\frac{1}{3}$

B $\frac{1}{2}$

C $\frac{3}{4}$

D $\frac{2}{3}$ ✓

WORKED SOLUTION & VERIFICATION

To find the probability that the chosen marble is NOT red, determine the number of favorable outcomes and divide by the total number of outcomes.

1. Find the total number of marbles in the bag:

$8 + 6 + 10 = 24$

2. Find the number of marbles that are not red (the green and blue marbles):

$6 + 10 = 16$

Alternatively, use the complement rule:

$P(\text{not red}) = 1 - P(\text{red}) = 1 - \frac{8}{24} = 1 - \frac{1}{3} = \frac{2}{3}$

3. Calculate the probability:

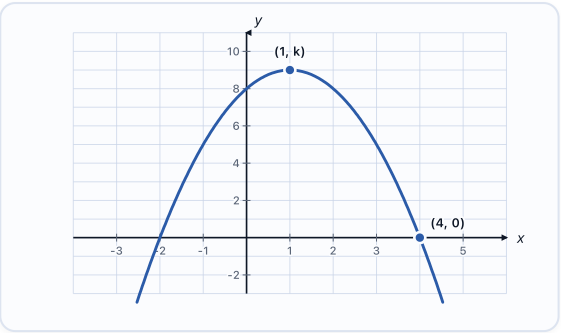
$P(\text{not red}) = \frac{16}{24} = \frac{2}{3}$

Common Misconceptions:

- $\frac{1}{3}$ is the probability of selecting a red marble ($\frac{8}{24}$), failing to account for the word NOT.
- $\frac{1}{2}$ or $\frac{3}{4}$ typically result from counting only one of the non-red colors or miscounting the total number of marbles.

12 MCQ

The graph of the quadratic function $f(x) = -x^2 + bx + c$ is shown in the standard (x, y) coordinate plane below. The vertex of the parabola is at $(1, k)$, and the graph passes through the point $(4, 0)$. What is the value of k?



A 9 ✓

B 6

C 8

D 12

WORKED SOLUTION & VERIFICATION

We can solve for k using two algebraic methods:

Method 1: Symmetry and Factored Form

The axis of symmetry of the parabola passes vertically through the vertex, so its equation is $x = 1$. Since $(4, 0)$ is an x-intercept, its horizontal distance from the axis of symmetry is $4 - 1 = 3$. By symmetry, the other x-intercept must lie 3 units to the left of $x = 1$, which is at $x = 1 - 3 = -2$.

Since the leading coefficient of $f(x) = -x^2 + bx + c$ is $a = -1$, we can write $f(x)$ in factored form:

$$f(x) = -(x - 4)(x - (-2)) = -(x - 4)(x + 2)$$

The value of k is the y-coordinate at the vertex $x = 1$:

$$k = f(1) = -(1 - 4)(1 + 2) = -(-3)(3) = 9$$

Method 2: Vertex Formula and Standard Form

For $f(x) = -x^2 + bx + c$, the x-coordinate of the vertex is given by:

$$x = -\frac{b}{2a} = -\frac{b}{2(-1)} = \frac{b}{2}$$

Setting this equal to 1 yields $b = 2$.

Since $(4, 0)$ lies on the parabola, substitute $x = 4$ and $y = 0$:

$$-(4)^2 + 2(4) + c = 0 \implies -16 + 8 + c = 0 \implies c = 8$$

Thus, the function is $f(x) = -x^2 + 2x + 8$. Finally, find $k = f(1)$:

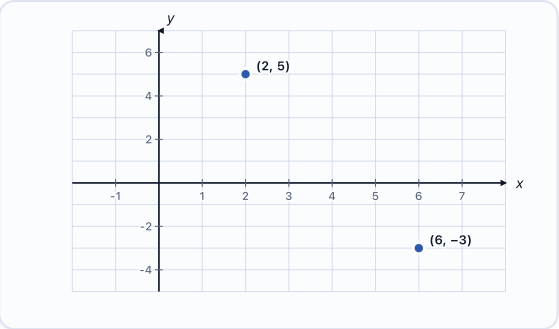
$$k = -(1)^2 + 2(1) + 8 = -1 + 2 + 8 = 9$$

Common Traps:

- Confusing the y-intercept with the vertex gives $c = 8$.
- Forgetting that the leading coefficient is -1 can cause sign errors during expansion.

13 MCQ

In the standard (x, y) coordinate plane, line l contains the points $(2, 5)$ and $(6, -3)$. Line k is perpendicular to the segment connecting these two points and passes through its midpoint. What is the y-intercept of line k?



A 9

B -1 ✓

C -6

D 3

WORKED SOLUTION & VERIFICATION

To find the y-intercept of line k, perform the following steps:

1. **Find the midpoint of the segment:**

$$M = \left(\frac{2+6}{2}, \frac{5+(-3)}{2}\right) = \left(\frac{8}{2}, \frac{2}{2}\right) = (4, 1)$$

2. **Find the slope of line l:**

$$m_l = \frac{-3-5}{6-2} = \frac{-8}{4} = -2$$

3. **Determine the slope of line k:**

Since line k is perpendicular to line l, its slope is the negative reciprocal of m_l:

$$m_k = -\frac{1}{-2} = \frac{1}{2}$$

4. **Find the y-intercept of line k:**

Using the slope-intercept form $y = m_k x + b$ with the midpoint $(4, 1)$:

$$1 = \frac{1}{2}(4) + b$$

$$1 = 2 + b \implies b = -1$$

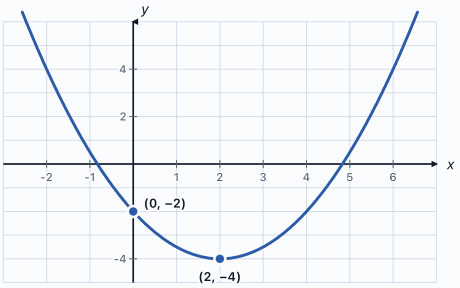
Therefore, the y-intercept of line k is -1.

Common Traps:

- Choice 9 comes from incorrectly using the original slope -2 instead of the negative reciprocal ($1 = -2(4) + b \implies b = 9$).
- Choice 3 comes from failing to negate the reciprocal slope, using $m = -\frac{1}{2}$ instead of $\frac{1}{2}$.
- Choice -6 comes from using the endpoint $(6, -3)$ rather than the midpoint $(4, 1)$.

14 MCQ

The graph of the quadratic function $f(x) = a(x-h)^2 + k$ is shown in the standard (x, y) coordinate plane below, with vertex $(2, -4)$ and y-intercept $(0, -2)$. If the point $(6, c)$ lies on the graph, what is the value of c?



- **A** 4 ✓
- **B** 2
- **C** 8
- **D** 12

WORKED SOLUTION & VERIFICATION

1. Identify the vertex form of the parabola:
Since the vertex is $(h, k) = (2, -4)$, the function is:

$$f(x) = a(x-2)^2 - 4$$

2. Solve for the leading parameter a using the given y-intercept $(0, -2)$:

$$-2 = a(0-2)^2 - 4$$

$$-2 = 4a - 4$$

$$2 = 4a \implies a = \frac{1}{2}$$

3. Determine the full equation of the function:

$$f(x) = \frac{1}{2}(x-2)^2 - 4$$

4. Find c by substituting $x = 6$:

$$c = f(6) = \frac{1}{2}(6-2)^2 - 4 = \frac{1}{2}(16) - 4 = 8 - 4 = 4$$

Misconception traps:
- Assuming $a = 1$ leads to $(6-2)^2 - 4 = 12$ (Choice D).
- Forgetting to subtract $k = 4$ leads to $\frac{1}{2}(16) = 8$ (Choice C).

15 MCQ

On the number line shown below, point A is located at -16 and point B is located at 24. Point P lies on the line segment connecting A and B such that the ratio of the length of $\overline{AP}$ to the length of $\overline{PB}$ is $3:2$. What is the coordinate of point P?



A 0

B 8 ✓

C 4

D 16

WORKED SOLUTION & VERIFICATION

To find the coordinate of point P, follow these steps:

1. Find the total distance between points A and B:

$$AB = 24 - (-16) = 24 + 16 = 40$$

2. Determine the fraction of the total distance represented by $\overline{AP}$. The ratio of the length of $\overline{AP}$ to $\overline{PB}$ is $3:2$, meaning there are $3 + 2 = 5$ equal parts in total. Thus, $\overline{AP}$ accounts for $\frac{3}{5}$ of the total distance:

$$AP = \frac{3}{5} \times 40 = 24$$

3. Add this distance to the starting coordinate of point A:

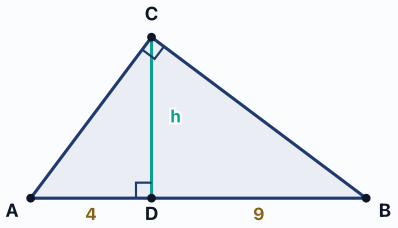
$$\text{Coordinate of } P = -16 + 24 = 8$$

Check against misconceptions:

- Finding the midpoint $\frac{-16+24}{2} = 4$ ignores the given $3:2$ ratio.
- Reversing the ratio to $2:3$ yields $-16 + \frac{2}{5}(40) = 0$.
- Calculating the length of $\overline{PB} = 16$ gives the remaining distance rather than the coordinate of P.

16 MCQ

In the figure below, $\triangle ABC$ is a right triangle with right angle at C, and altitude $\overline{CD}$ of length h is perpendicular to hypotenuse $\overline{AB}$. If $AD = 4$ and $DB = 9$, what is the area of $\triangle ABC$?



A 27

B 36

C 39 ✓

D 78

WORKED SOLUTION & VERIFICATION

To find the area of $\triangle ABC$, we need the base AB and the corresponding altitude $CD = h$.

1. Find the altitude h:

By the Geometric Mean (Altitude) Theorem for right triangles, the altitude to the hypotenuse is the geometric mean of the two segments of the hypotenuse:

$$h^2 = AD \cdot DB = 4 \cdot 9 = 36$$

Taking the square root gives:

$$h = \sqrt{36} = 6$$

2. Find the total length of base $\overline{AB}$:

$$AB = AD + DB = 4 + 9 = 13$$

3. Calculate the area of $\triangle ABC$:

$$\text{Area} = \frac{1}{2} \cdot \text{base} \cdot \text{height} = \frac{1}{2} \cdot 13 \cdot 6 = 39$$

Common traps:

- Finding the area of only sub-triangle $\triangle CDB$: $\frac{1}{2} \cdot 9 \cdot 6 = 27$.
- Confusing the altitude squared with the area: $4 \cdot 9 = 36$.
- Forgetting the factor of $\frac{1}{2}$ in the area formula: $13 \cdot 6 = 78$.

17 MCQ

A list consists of 7 distinct integers. The arithmetic mean of the 7 integers is 24, and the median is 22. If the 3 smallest integers are 15, 18, and 20, what is the maximum possible value of the largest integer in the list?

A 47

B 48

C 49

D **46** ✓

WORKED SOLUTION & VERIFICATION

Let the 7 distinct integers in ascending order be $x_1 < x_2 < x_3 < x_4 < x_5 < x_6 < x_7$.

1. From the problem, the sum of all 7 integers is:

$$\text{Sum} = 7 \times 24 = 168$$

2. The median of a set of 7 ordered integers is the 4th integer, so $x_4 = 22$.

3. The 3 smallest integers are given as $x_1 = 15$, $x_2 = 18$, and $x_3 = 20$.

4. The sum of the first four integers is:

$$15 + 18 + 20 + 22 = 75$$

5. The sum of the remaining three integers must be:

$$x_5 + x_6 + x_7 = 168 - 75 = 93$$

6. To maximize x_7, we must minimize x_5 and x_6. Since all integers in the list are distinct and greater than $x_4 = 22$, the smallest possible integer values are:

$$x_5 = 23 \quad \text{and} \quad x_6 = 24$$

7. Substituting these minimum values gives the maximum for x_7:

$$x_7 = 93 - (23 + 24) = 93 - 47 = 46$$

Common Misconceptions:

- Allowing repeated values (forgetting the integers must be distinct) yields $x_5 = 22$ and $x_6 = 22$, resulting in $93 - 44 = 49$.
- Allowing $x_5 = 22$ and $x_6 = 23$ yields $93 - 45 = 48$.

18 MCQ

The function $f(x) = a\sin(bx) + c$ has a maximum value of 9, a minimum value of -1, and a period of $\frac{\pi}{2}$. If $a > 0$ and $b > 0$, what is the value of $a + b + c$?

A 9

B 13 ✓

C 11

D 18

WORKED SOLUTION & VERIFICATION

To find the value of $a + b + c$, determine each parameter from the given properties of the sinusoidal function:

1. **Amplitude a and Vertical Shift c:**

Since $-1 \le \sin(bx) \le 1$ and $a > 0$, the maximum and minimum values of $f(x)$ are:

$$\text{Maximum} = c + a = 9$$

$$\text{Minimum} = c - a = -1$$

Adding these two equations gives:

$$2c = 8 \implies c = 4$$

Substituting $c = 4$ into $c + a = 9$:

$$4 + a = 9 \implies a = 5$$

2. **Frequency Parameter b:**

The period T of a sine function $\sin(bx)$ with $b > 0$ is given by $T = \frac{2\pi}{b}$. Setting this equal to the given period $\frac{\pi}{2}$:

$$\frac{2\pi}{b} = \frac{\pi}{2} \implies b\pi = 4\pi \implies b = 4$$

3. **Sum of the Parameters:**

$$a + b + c = 5 + 4 + 4 = 13$$

Common Traps:

- Choice A (9) mistakenly calculates only $a + c = 9$, ignoring the parameter b.
- Choice B (11) mistakenly uses the tangent period formula $\frac{\pi}{b}$, yielding $b = 2$ and $5 + 2 + 4 = 11$.
- Choice D (18) confuses the amplitude with the full vertical span $9 - (-1) = 10$, leading to $10 + 4 + 4 = 18$.

19 MCQ

The quadratic function $f(x) = x^2 - 2kx + (3k + 10)$ has 2 distinct real zeros. If both zeros are strictly greater than 2, how many integer values of k are possible?

A 7

B 9

C 11

D 8 ✓

WORKED SOLUTION & VERIFICATION

To have both zeros of the upward-opening parabola $f(x) = x^2 - 2kx + (3k + 10)$ be real, distinct, and strictly greater than 2, three conditions must hold simultaneously:

1. **Positive Discriminant ($\Delta > 0$):**

$\Delta = (-2k)^2 - 4(1)(3k + 10) = 4k^2 - 12k - 40 = 4(k - 5)(k + 2) > 0.$

This requires $k > 5$ or $k < -2$.

2. **Vertex Position ($x_v > 2$):**

The axis of symmetry of $f(x)$ is at $x = -\frac{-2k}{2(1)} = k$. For both zeros to lie to the right of 2, the midpoint of the zeros must satisfy:

$k > 2.$

Intersecting $k > 2$ with $k > 5$ or $k < -2$ narrows the condition to $k > 5$.

3. **Boundary Value ($f(2) > 0$):**

Since the parabola opens upward, having 2 to the left of both zeros requires:

$f(2) = 2^2 - 2k(2) + 3k + 10 = 14 - k > 0 \implies k < 14.$

Combining all inequalities yields:

$5 < k < 14.$

The possible integer values for k are $\{6, 7, 8, 9, 10, 11, 12, 13\}$.

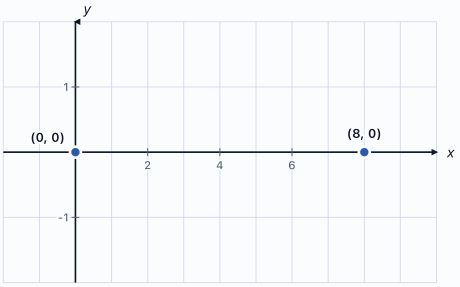
Counting these integers gives $13 - 6 + 1 = 8.$

Common Misconceptions:

- Forgetting the discriminant restriction $\Delta > 0$ leads to $2 < k < 14$, which falsely gives 11 integers.
- Including $k = 5$ (where the zeros are not distinct) or $k = 14$ (where one zero is exactly 2) leads to 9 integers.
- An off-by-one counting error may produce 7.

20 MCQ

In the standard (x, y) coordinate plane, a triangle has vertices at $O(0, 0)$, $A(8, 0)$, and $B(2, 6)$. A horizontal line with equation $y = c$, where $0 < c < 6$, divides the interior of $\triangle OAB$ into two regions of equal area. What is the value of c?



A $6 - 2\sqrt{3}$

B 3

C $3\sqrt{2}$

D $6 - 3\sqrt{2}$ ✓

WORKED SOLUTION & VERIFICATION

To find the value of c, we first determine the total area of $\triangle OAB$. The base along the x-axis lies on the segment from $O(0, 0)$ to $A(8, 0)$, which has length $b = 8$. The height of the triangle corresponds to the y-coordinate of vertex $B(2, 6)$, so $h = 6$.

$$\text{Area}(\triangle OAB) = \frac{1}{2} \times 8 \times 6 = 24$$

The horizontal line $y = c$ is parallel to the base $\overline{OA}$ (which lies on the line $y = 0$). A line parallel to one side of a triangle creates an upper triangle that is similar to the original triangle $\triangle OAB$.
Since the line $y = c$ divides $\triangle OAB$ into two regions of equal area, the upper triangle with vertex at $B(2, 6)$ must have an area equal to half of the total area:

$$\text{Area(upper)} = \frac{24}{2} = 12$$

For similar geometric figures, the ratio of their areas is equal to the square of the ratio of their corresponding linear dimensions (scale factor k):

$$\frac{\text{Area(upper)}}{\text{Area}(\triangle OAB)} = \left(\frac{h_{\text{upper}}}{h}\right)^2 = \frac{12}{24} = \frac{1}{2}$$

Taking the square root gives the ratio of the heights:

$$\frac{h_{\text{upper}}}{6} = \frac{1}{\sqrt{2}} \implies h_{\text{upper}} = \frac{6}{\sqrt{2}} = 3\sqrt{2}$$

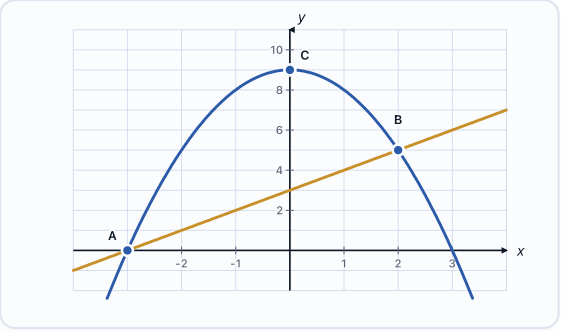
The height of the upper triangle is the vertical distance from $B(2, 6)$ down to the horizontal line $y = c$:

$$h_{\text{upper}} = 6 - c = 3\sqrt{2} \implies c = 6 - 3\sqrt{2}$$

Common misconceptions:
- Choosing 3 assumes the line bisecting the area simply bisects the height, ignoring that the width narrows towards the top vertex.
- Choosing $3\sqrt{2}$ confuses the height of the upper triangle with the y-intercept c of the line.
- Choosing $6 - 2\sqrt{3}$ results from incorrectly equating $(6 - c)^2$ to 12 instead of 18.

21 MCQ

In the standard (x, y) coordinate plane below, the graph of the quadratic function $f(x) = 9 - x^2$ and the line $g(x) = x + 3$ intersect at points A and B, where A has a negative x-coordinate and B has a positive x-coordinate. Point C is the vertex of the parabola $y = f(x)$. What is the area, in square coordinate units, of $\triangle ABC$?



A 12.5

B 15 ✓

C 22.5

D 30

WORKED SOLUTION & VERIFICATION

To find the area of $\triangle ABC$, we first determine the coordinates of its three vertices: A, B, and C.

1. **Vertex C**:

The function $f(x) = 9 - x^2$ is a downward-opening parabola with its maximum at $x = 0$.

$f(0) = 9 - 0^2 = 9$, so the vertex is $C(0, 9)$.

2. **Intersection Points A and B**:

Set $f(x) = g(x)$:

$$9 - x^2 = x + 3$$

Rearrange into standard quadratic form:

$$x^2 + x - 6 = 0$$

Factor the quadratic equation:

$$(x + 3)(x - 2) = 0$$

Thus, the x-coordinates of the intersection points are $x = -3$ and $x = 2$.

- For $x = -3$: $y = g(-3) = -3 + 3 = 0$, giving point $A(-3, 0)$.
- For $x = 2$: $y = g(2) = 2 + 3 = 5$, giving point $B(2, 5)$.

3. **Area of $\triangle ABC$**:

Using the coordinate geometry area formula (Shoelace formula) for vertices $(-3, 0)$, $(2, 5)$, and $(0, 9)$:

$$\text{Area} = \frac{1}{2}|x_A(y_B - y_C) + x_B(y_C - y_A) + x_C(y_A - y_B)|$$

Substitute the coordinates:

$$\text{Area} = \frac{1}{2}|-3(5 - 9) + 2(9 - 0) + 0(0 - 5)|$$

$$\text{Area} = \frac{1}{2}|-3(-4) + 2(9) + 0| = \frac{1}{2}|12 + 18| = \frac{1}{2}(30) = 15$$

Alternatively, enclosing $\triangle ABC$ in a bounding rectangle with horizontal span from $x = -3$ to $x = 2$ (width 5) and vertical span from $y = 0$ to $y = 9$ (height 9) gives a total box area of $5 \times 9 = 45$. Subtracting the three surrounding right triangles gives $45 - \frac{1}{2}(5)(5) - \frac{1}{2}(2)(4) - \frac{1}{2}(3)(9) = 45 - 12.5 - 4 - 13.5 = 15$.

> Common errors include forgetting the factor of $\frac{1}{2}$ (yielding 30) or multiplying the horizontal width by the total height and dividing by 2 without adjusting for the non-right shape (yielding 22.5).

22 MCQ

Every student in the graduating class of Oakridge High School plans to do exactly one of the following after graduation: attend a 2-year college, attend a 4-year university, or enter the workforce directly.

• The ratio of the number of students planning to attend a 2-year college to the number of students planning to attend a 4-year university is $3:5$.

• Of the students planning to attend a 2-year college, exactly 80% will attend an in-state college.

• Of the students planning to attend a 4-year university, exactly 72% will attend an in-state university.

• The total number of students planning to attend an in-state college or university accounts for exactly 45% of the entire graduating class.

If the number of students in every category and subcategory must be a positive integer, what is the minimum possible number of students in the graduating class?

A 200 ✓	B 100
C 120	D 40

WORKED SOLUTION & VERIFICATION

Let N be the total number of students in the graduating class.

1. **Express college counts in terms of a multiplier x:**

The ratio of students attending a 2-year college to those attending a 4-year university is $3:5$. Thus, for some positive value x:

$$C_2 = 3x \quad \text{and} \quad C_4 = 5x$$

Since both $3x$ and $5x$ must be integers, and $\gcd(3,5)=1$, $x = 2(5x) - 3(3x)$ must be a positive integer.

2. **Determine the number of in-state students:**

- In-state 2-year college students: $0.80(3x) = 2.4x = \frac{12}{5}x$.
- In-state 4-year university students: $0.72(5x) = 3.6x = \frac{18}{5}x$.

For both quantities to be integers, 5 must divide x, meaning x must be a multiple of 5.

3. **Relate in-state students to the total class size N:**

The total number of in-state students is:

$$2.4x + 3.6x = 6x$$

We are given that this total is 45% of N:

$$6x = 0.45N = \frac{9}{20}N \implies N = \frac{20}{9}(6x) = \frac{40}{3}x$$

4. **Apply integer constraints:**

For $N = \frac{40}{3}x$ to be an integer, 3 must divide x (since $\gcd(40,3)=1$).

Since x must be divisible by both 5 and 3, and $\gcd(5,3)=1$, x must be a multiple of:

$$\text{lcm}(3,5) = 15$$

5. **Find the minimum class size:**

The smallest positive integer value for x is $x = 15$. Substituting $x = 15$ into the expression for N:

$$N = \frac{40}{3}(15) = 40 \times 5 = 200$$

Check of subcategories for $x = 15$:

- 2-year college: 45 (36 in-state, 9 out-of-state)
- 4-year university: 75 (54 in-state, 21 out-of-state)
- In-state total: $36 + 54 = 90$, which is $\frac{90}{200} = 45\%$

- Workforce: $200 - (45 + 75) = 80$

All counts are positive integers. The minimum possible number of students is 200.

Common Misconceptions:

- Choice A (40): Results from setting $x = 3$ to cancel the denominator of 3, ignoring the requirement that x must also be a multiple of 5 (which would yield 7.2 in-state 2-year students).
- Choice B (100): Results from assuming a baseline of 100 students simply because percentages are used, which yields fractional student counts (e.g., $x = 7.5$ and 22.5 students in 2-year colleges).
- Choice C (120): Mistaking the total number of college-bound students ($45 + 75 = 120$) for the entire graduating class.

23 MCQ

An isosceles trapezoid has parallel bases of lengths 4 cm and 16 cm. A circle is inscribed inside the trapezoid such that it is tangent to all four sides. What is the area, in square centimeters, of the region inside the trapezoid that lies outside the circle?

A $64 - 16\pi$

B $80 - 16\pi$ ✓

C $80 - 25\pi$

D $100 - 25\pi$

WORKED SOLUTION & VERIFICATION

1. **Determine the leg lengths of the trapezoid:**

Let the trapezoid be $ABCD$ with parallel bases $AB = 4$ cm and $CD = 16$ cm, and nonparallel legs $AD = BC$. Because a circle is tangent to all four sides, the trapezoid is a tangential quadrilateral. Tangent segments drawn from any vertex to the circle are equal in length. By symmetry, the points of tangency on the bases bisect each base:

- The segments from the top vertices to the top tangency point each measure $4/2 = 2$ cm.
- The segments from the bottom vertices to the bottom tangency point each measure $16/2 = 8$ cm.

Therefore, each leg has length:

$$AD = BC = 2 + 8 = 10 \text{ cm}$$

2. **Find the height of the trapezoid:**

Drop perpendicular altitudes from the endpoints of the shorter base to the longer base. The projection of each slanted leg onto the longer base has length:

$$\frac{16 - 4}{2} = 6 \text{ cm}$$

By the Pythagorean theorem on the resulting right triangle:

$$h^2 + 6^2 = 10^2 \implies h^2 + 36 = 100 \implies h^2 = 64 \implies h = 8 \text{ cm}$$

3. **Calculate the radius of the inscribed circle:**

The distance between the horizontal parallel tangent lines is the diameter of the circle, so:

$$2r = h = 8 \text{ cm} \implies r = 4 \text{ cm}$$

4. **Calculate the required area:**

- Area of the trapezoid:

$$\text{Area}_{\text{trap}} = \frac{4 + 16}{2} \times 8 = 10 \times 8 = 80 \text{ cm}^2$$

- Area of the inscribed circle:

$$\text{Area}_{\text{circle}} = \pi r^2 = \pi(4^2) = 16\pi \text{ cm}^2$$

- Area of the region outside the circle:

$$\text{Area} = 80 - 16\pi \text{ cm}^2$$

Common Misconceptions:

- Misconception 1 (Choice B): Assuming the circle's diameter is the average of the bases (10), which would incorrectly give $r = 5$ and an area of $80 - 25\pi$.
- Misconception 2 (Choice C): Conflating the leg length with the perpendicular height $(h = 10)$, yielding a trapezoid area of 100 and circle area of 25π.

24 MCQ

A box contains 10 cards, each labeled with a distinct integer from 1 through 10. If 3 cards are drawn at random from the box without replacement, what is the probability that the sum of the numbers on the 3 drawn cards is divisible by 3?

- **A** $\frac{7}{20}$ ✓
- **B** $\frac{1}{4}$
- **C** $\frac{3}{10}$
- **D** $\frac{1}{3}$

WORKED SOLUTION & VERIFICATION

To find the probability that the sum of the 3 chosen integers is divisible by 3, we synthesize combinatorial counting with modular arithmetic.

1. **Total Number of Outcomes:**

The total number of ways to choose 3 cards from 10 without replacement is given by the combination formula:

$$\binom{10}{3} = \frac{10 \times 9 \times 8}{3 \times 2 \times 1} = 120$$

2. **Partition by Remainder modulo 3:**

Group the integers 1 through 10 by their remainder when divided by 3:

- **Remainder 0:** $\{3, 6, 9\}$ (3 cards)
- **Remainder 1:** $\{1, 4, 7, 10\}$ (4 cards)
- **Remainder 2:** $\{2, 5, 8\}$ (3 cards)

3. **Favorable Combinations:**

The sum of three integers is divisible by 3 if and only if the sum of their remainders modulo 3 is a multiple of 3. There are only two ways this can occur:

- **Subcase A: All 3 cards have the same remainder modulo 3:**
- Three cards with remainder 0: $\binom{3}{3} = 1$
- Three cards with remainder 1: $\binom{4}{3} = 4$
- Three cards with remainder 2: $\binom{3}{3} = 1$

Total for this subcase: $1 + 4 + 1 = 6$ outcomes.

- **Subcase B: Exactly one card from each remainder group (remainders 0, 1, and 2):**
- $\binom{3}{1} \times \binom{4}{1} \times \binom{3}{1} = 3 \times 4 \times 3 = 36$ outcomes.

4. **Total Favorable Outcomes and Probability:**

Adding all favorable outcomes yields:

$$6 + 36 = 42$$

Therefore, the probability is:

$$P = \frac{42}{120} = \frac{7}{20}$$

Common Misconceptions:

- **Guessing $\frac{1}{3}$:** It is tempting to assume that sums are equally distributed among the three possible remainders modulo 3. However, because the remainder groups do not have equal cardinality (remainder 1 has 4 members while the others have 3) and sampling is without replacement, the distribution is not uniform.
- **Obtaining $\frac{3}{10}$:** This occurs if one considers only Subcase B (one of each remainder, $36/120$) and overlooks the combinations where all three cards share the same remainder.

25 MCQ

In $\triangle ABC$, side AB has length 6, side AC has length 10, and the measure of $\angle BAC$ is 120°. The angle bisector of $\angle BAC$ intersects side BC at point D. What is the length of segment AD?

A $\frac{30}{7}$

B $\frac{15}{4}$ ✓

C $\frac{15\sqrt{3}}{4}$

D $\frac{15}{2}$

WORKED SOLUTION & VERIFICATION

We can find the length of the angle bisector AD by decomposing the area of $\triangle ABC$ into the sum of the areas of $\triangle ABD$ and $\triangle ACD$.

Since AD bisects $\angle BAC = 120^\circ$, we have:

$$\angle BAD = \angle CAD = 60^\circ$$

Using the sine formula for the area of a triangle, $\text{Area} = \frac{1}{2}ab\sin\theta$:

$$\text{Area}(\triangle ABC) = \frac{1}{2} \cdot AB \cdot AC \cdot \sin(120^\circ) = \frac{1}{2} \cdot 6 \cdot 10 \cdot \frac{\sqrt{3}}{2} = 15\sqrt{3}$$

The sum of the areas of the two smaller triangles is:

$$\text{Area}(\triangle ABD) = \frac{1}{2} \cdot AB \cdot AD \cdot \sin(60^\circ) = \frac{1}{2} \cdot 6 \cdot AD \cdot \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2}AD$$

$$\text{Area}(\triangle ACD) = \frac{1}{2} \cdot AC \cdot AD \cdot \sin(60^\circ) = \frac{1}{2} \cdot 10 \cdot AD \cdot \frac{\sqrt{3}}{2} = \frac{5\sqrt{3}}{2}AD$$

Setting the total area equal to the sum of the two parts:

$$\frac{3\sqrt{3}}{2}AD + \frac{5\sqrt{3}}{2}AD = 15\sqrt{3}$$

$$4\sqrt{3}AD = 15\sqrt{3}$$

$$AD = \frac{15}{4}$$

Common Misconceptions:

- Choice B ($\frac{15\sqrt{3}}{4}$) results from incorrectly multiplying by $\sin(60^\circ)$ again when solving for AD.
- Choice C ($\frac{15}{2}$) is the harmonic mean $\frac{2bc}{b+c}$, forgetting to multiply by $\cos(60^\circ) = \frac{1}{2}$.
- Choice D ($\frac{30}{7}$) comes from using $\frac{AB \cdot AC}{BC}$ with $BC = 14$, confusing the altitude formula with the angle bisector.